GLUCOTWIN: SYSTEM ARCHITECTURE & DATA FLOW

Type 2 Diabetes Digital Twin for 120-Minute Adverse Glycemic Excursion Prediction | DTC 2026

EHR Demographics

Synthea / FHIR R4
101M India T2D Profile
HbA1c, eGFR, BMI, Meds

Continuous Glucose (CGM)

Ohio T1DM + AR(1) Sim
5-Min Sampling [40-400 mg/dL]
MARD 7% Sensor Noise

Wearable Autonomic Biometrics

HRV (RMSSD, SDNN, LF/HF)
Steps & Metabolic Activity
Parasympathetic Suppression

Data Ingestion Streamer

WearableStreamManager
Data Quality & Gap Padding
ClinicalValidator Filter

89-Feature Engineering

Static EHR (23) + CGM Stats (20)
Clinical Indices (12: TIR, MAGE)
HRV Autonomic Coupling (12)

PatientTwin State Machine

Circular Buffers (288 steps / 24h)
eAG Baseline = 28.7 * HbA1c - 46.7
Velocity Bands (5 Trend States)

Pharmacokinetic IOB Model

Hovorka Biexponential Model
Active Insulin-on-Board
Rule of 1800 ISF Sensitivity

Multi-Patient TwinEngine

Concurrent Twin Registry
Async State Persistence (SQLite)
Fatigue-Suppressed Alerts

XGBoost Tabular Predictor

89 Engineered Features
scale_pos_weight = 5.7
Tree Hist Method (AUC 0.91)

PyTorch Temporal LSTM

48-Timestep Causal Window
(Batch, 48, 3 Channels)
Monte Carlo Dropout 95% CI

Adaptive Ensemble & SHAP

Insulin: 35% XGB / 65% LSTM
Non-Insulin: 60% XGB / 40% LSTM
Top 10 SHAP Explanations

FastAPI & Clinician Cockpit

Sub-200ms REST Endpoints
Live WebSocket Stream (/ws)
Doctor Twin Cockpit UI

Key Metrics: AUC-ROC: 0.91 | Adverse Event Precision: 87% | Latency: < 200ms | 120-Minute Early Intervention Window | DPDP Act 2023 Compliant